

```
static void __exit pen_exit(void)
{
    /* Deregister this driver with the USB subsystem */
    usb_deregister(&pen_driver);
}

module_init(pen_init);
module_exit(pen_exit);

MODULE_LICENSE("GPL");
MODULE_AUTHOR("Anil Kumar Pugalia <email@sarika-pugs.com>");
MODULE_DESCRIPTION("USB Pen Device Driver");
```

As a reminder, the usual steps for any Linux device driver may be repeated with the above code, along with the following steps for the pen drive:

- Build the driver (*pen_driver.ko*) by running *make*.
- Load the driver using *insmod pen_driver.ko*.
- Plug in the pen drive (after making sure that the *usb-storage* driver is not already loaded).
- Check for the dynamic creation of */dev/pen0* (0 being the minor number obtained—check *dmesg* logs for the value on your system).
- Possibly try some write/read on */dev/pen0* (you most

likely will get a connection time-out and/or broken pipe errors, because of non-conforming SCSI commands).

- Unplug the pen drive and look for */dev/pen0* to be gone.
- Unload the driver using *rmmdev pen_driver*.

Meanwhile, Pugs hooked up his first-of-its-kind creation—the Linux device driver kit (LDDK)—into his system for a live demonstration of the USB data transfers. "Aha! Finally a cool complete working USB driver," quipped Shweta, excited. "Want to have more fun? We could do a block driver over it..." added Pugs. "Oh! Really?" asked Shweta, with glee. "Yes. But before that, we need to understand the partitioning mechanisms," commented Pugs. **END** 

By: Anil Kumar Pugalia

The author is a freelance trainer in Linux internals, Linux device drivers, embedded Linux and related topics. Prior to this, he was at Intel and Nvidia. He has been working with Linux since 1994. A gold medalist from the Indian Institute of Science, Linux and knowledge sharing are two of his many passions. Creating and playing with open source hardware is one of his hobbies. Learn more about him at <http://profession.sarika-pugs.com/>, including information about his open source hardware freak-outs and his various Linux related training sessions and workshops. He can be reached at email@sarika-pugs.com.






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Go Online to learn Linux Device Drivers



Core skills like kernel programming or Linux device drivers should be learned at a slow and steady pace for fully comprehending it, says Mr. Raghu Bharadwaj, a lead trainer on core Linux programming. Unlike normal application software learning where it is mostly restricted to functionality based learning, system programming involves deep understanding and analysis of issues like the Linux kernel and its subsystems. Your options to learn these skills are classroom based training, learning through experience, self learning through trial and error and through an effective online course.

While all other modes of training has its own merits and demerits, online learning is definitely a step ahead, especially for courses like this. Online courses gives you the comfort of accessing it 24x7 and lets you learn things during hours of focus and interest. Another important aspect is that it permits you to go through each session any number of times and fully understand each and every topic effectively.

Mr. Raghu Bharadwaj is a lead trainer at Veda Solutions, you can listen to his **training videos** on Linux kernel and device drivers at www.techveda.org

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